Tony Chen

Los Angeles, CA • 412-944-8402 • tonychen1@ucla.edu • tonychen.cc

EDUCATION

University of California, Los Angeles *Expected Jun. 2030*
B.S., Electrical Engineering

Shady Side Academy, Pittsburgh, PA *Sep. 2022 – Jun. 2026*
- GPA 3.96/4.00. National Merit Finalist (Pennsylvania). ACT 36 composite; SAT 1560.
- AP Calculus BC (5), AP Computer Science A (5), AP Physics C: Mechanics (4), AP Physics C: E&M (4); coursework in linear algebra, differential equations, multivariable calculus, and calculus-based physics.

EXPERIENCE

NASA, Embedded Systems Engineering Intern *Summer 2025*
- Developed an autonomous mapping and placement pipeline in ROS2 on a rover platform.
- Built a LIDAR detection system to map an unknown area and generate a spatial model.
- Applied OpenCV and AprilTag detection to identify and place areas within the newly mapped space.

ENGINEERING PROJECTS

Automated Tracking Telescope *Jan. – Jun. 2025*
- Locates and tracks deep-sky objects using two stepper motors on independent axes.
- Camera attachment for long-exposure astrophotography, plus swappable eyepieces for variable magnification.
- Shutter system for brightness control; target magnification 200×, enough to resolve Jovian bands and Saturn's rings.

Custom Remote-Controlled Race Car *Oct. 2024 – Jan. 2025*
- 3D-printed chassis with a dual-motor drivetrain, independent electronic speed controllers, and a shared 5200 mAh 11.1 V LiPo pack.
- Ackermann steering geometry so the inner and outer wheels track at different optimal angles, tightening the turning radius and keeping the tires from scrubbing in corners.
- 15° caster angle to generate self-aligning torque, improving straight-line stability at speed and steering return after turns.

Electric Vehicle *Oct. 2024 – Apr. 2025*
- Science Olympiad event: Arduino-controlled vehicle that runs a track as fast as possible and stops at a target point, scored on combined time and stopping distance. The 7–10 m target distance is announced only after impound, so the vehicle had to be recalibrated to an unknown distance and run twice inside an 8-minute window.

EXTRACURRICULAR ACTIVITIES

Captain, Robotics Club, Shady Side Academy *Oct. 2024 – Jun. 2026*
- FIRST Tech Challenge: recruited, organized the team, led fundraising, and directed design, build, programming, and competition.
- Modeled the complete robot chassis from scratch in Fusion 360 using goBILDA components, adding a formal CAD phase the team had never had before, and wrote the control code in Java.

Computer Science Departmental Associate, Shady Side Academy *Oct. 2024 – Jun. 2026*
- One of five students selected to represent the CS department at school events, tutor peers, and promote computer science programs.

Writing Center English Tutor, Shady Side Academy *Jan. 2025 – Jun. 2026*
- Admitted to the Theory and Process of Writing course and selected to serve as a writing center scribe.

Science Olympiad, Shady Side Academy *Oct. 2024 – Jun. 2026*
- Competed in Electric Vehicle, Towers, Bungee, and miscellaneous study events.

Music and Athletics *2014 – 2026*
- First violin, Pittsburgh Youth Philharmonic Orchestra (2019–2026); ABRSM Piano Grade 8 and Theory Grade 5, both with Distinction; JV soccer, basketball, and track & field.

TECHNICAL SKILLS

Python, Java, C++, ROS2, OpenCV, Autodesk Fusion 360, FDM 3D printing, CNC milling, mechanical assembly, Motor drive, electronic speed controllers, stepper motor control, LiPo power systems